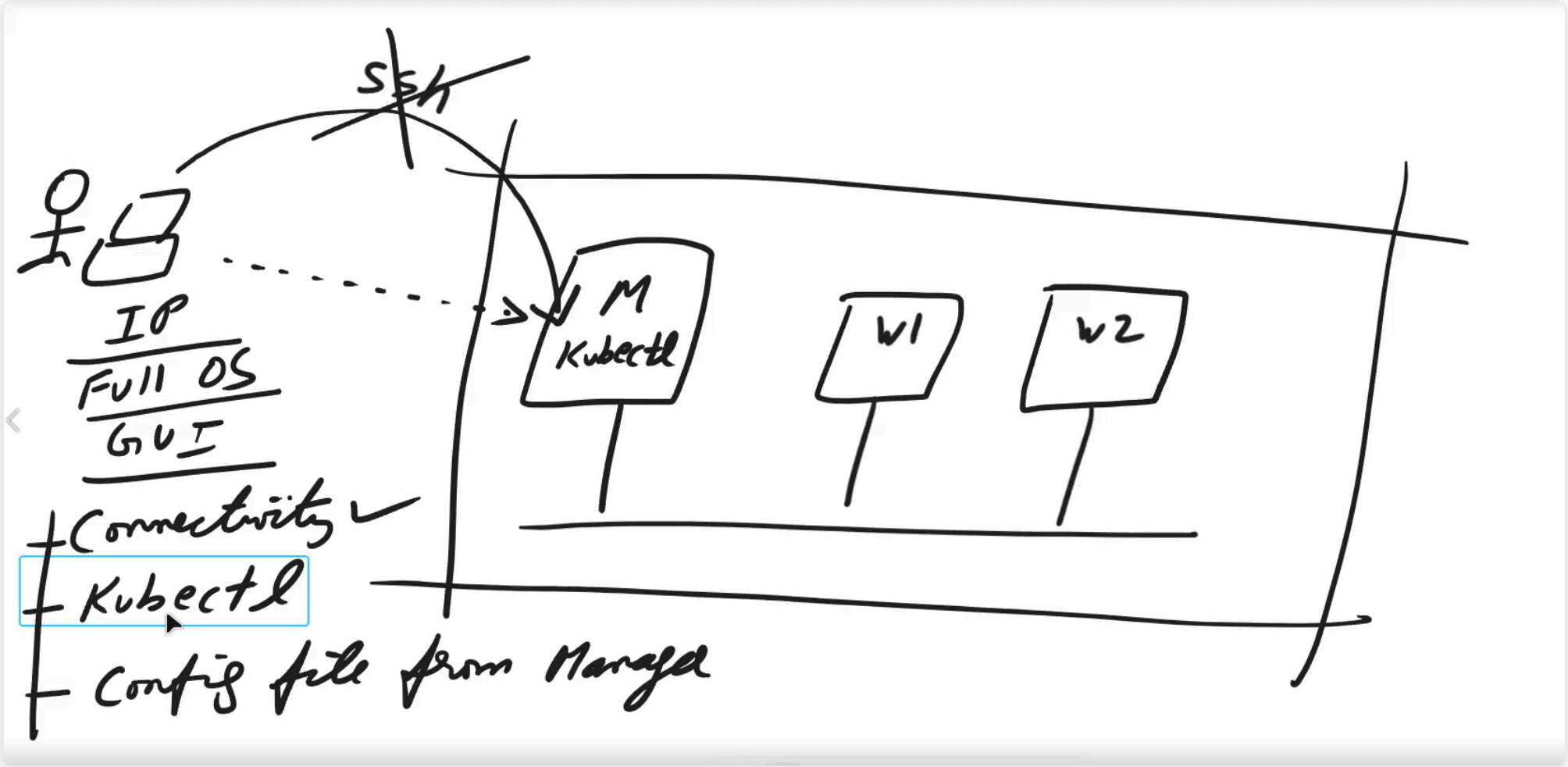
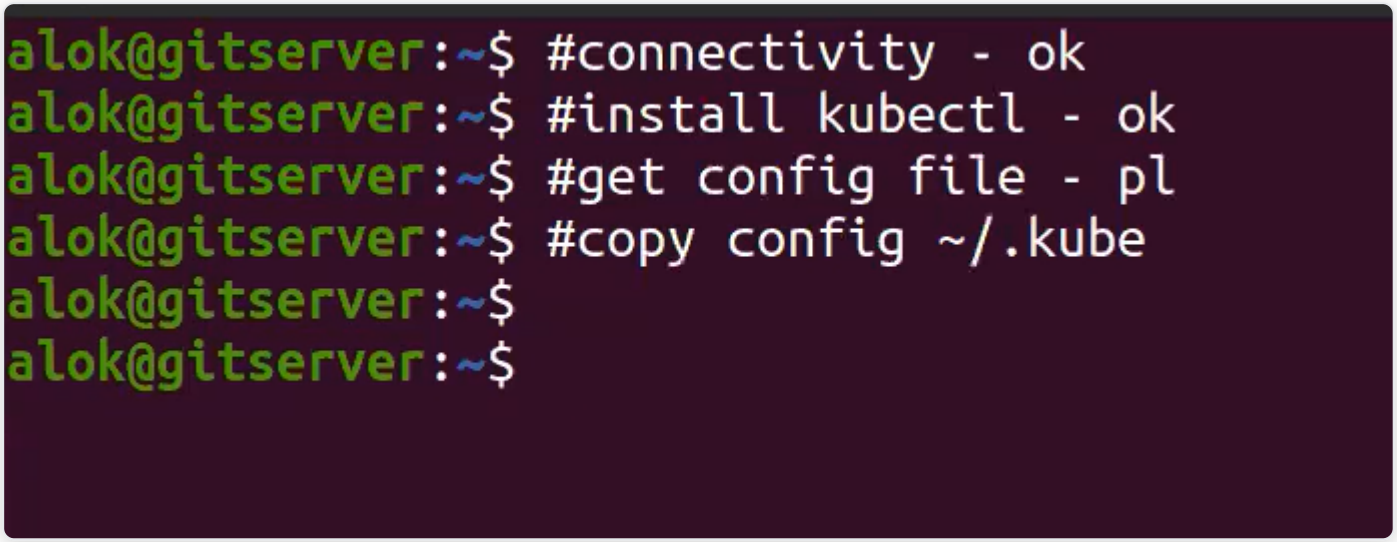
KUBERNETES CKA PREP › MY LEARNING NOTES › SESSION 03

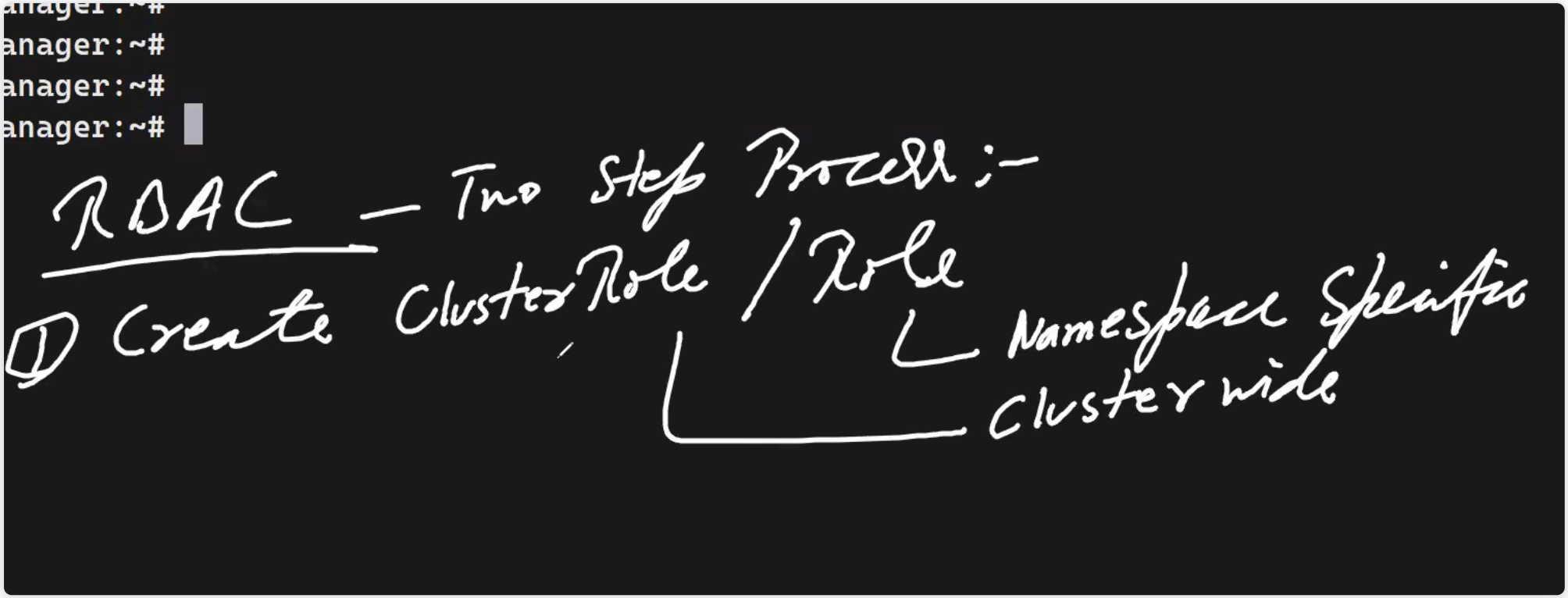
Screenshots: Production Access & RBAC Onboarding

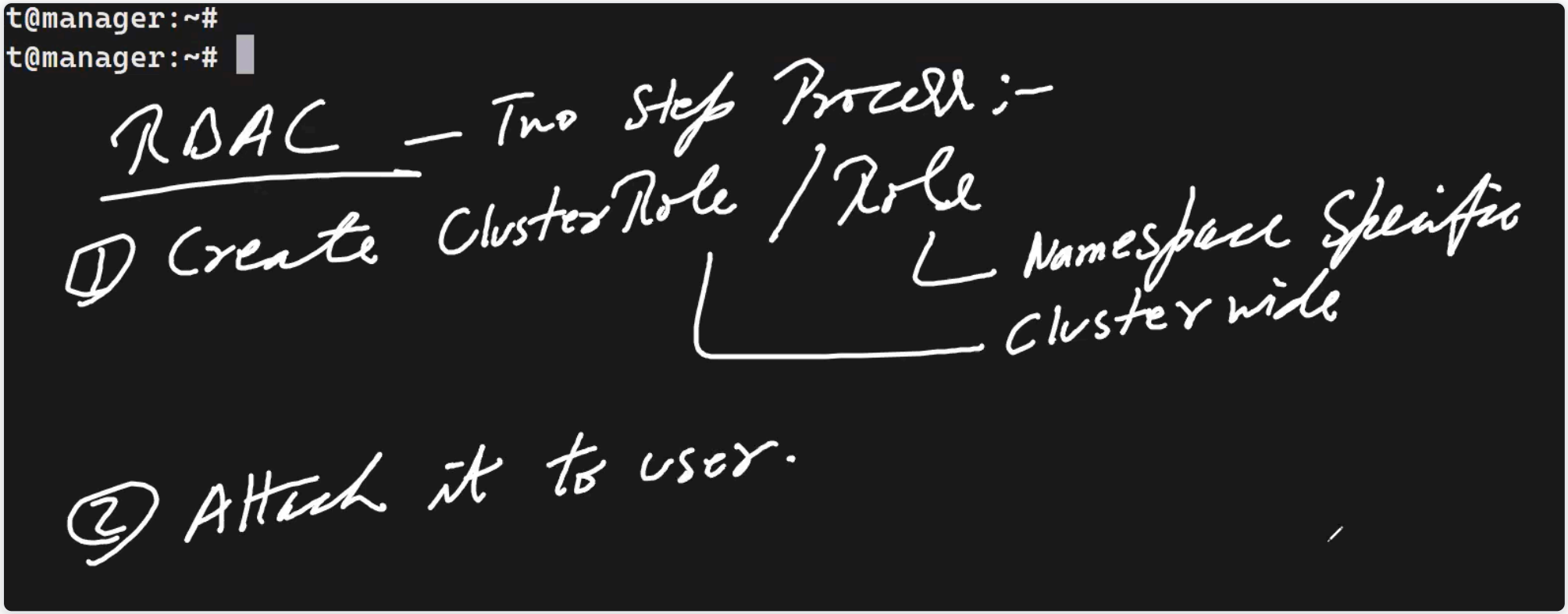
Raw screenshots captured while watching this session, kept alongside the written notes (03-production-access-and-rbac.md) for visual reference during interview revision.

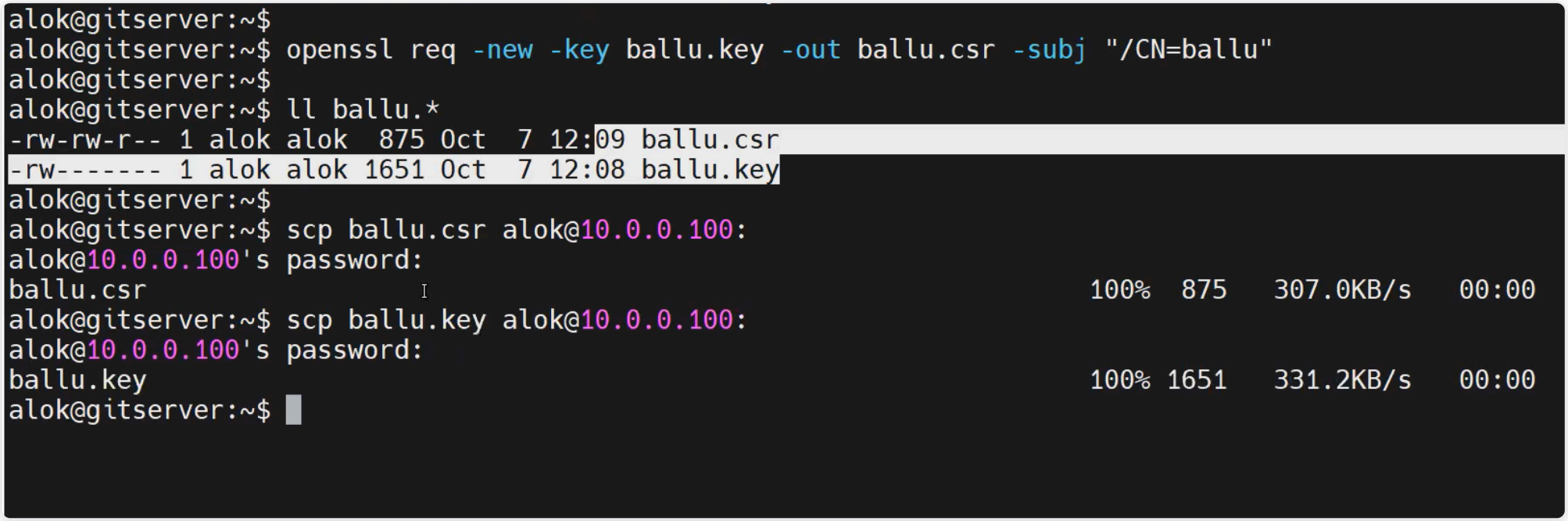
21 screenshots · captured 2026-09-11, 12:08 AM - 1:34 AM


ssh
IP
Full OS
GUI
Connectivity
Kubectl
Config file from Master
M
Kubectl
W1
W2


alok@gitserver:~$ #connectivity - ok
alok@gitserver:~$ #install kubectl - ok
alok@gitserver:~$ #get config file - pl
alok@gitserver:~$ #copy config ~/.kube
alok@gitserver:~$
alok@gitserver:~$


anager:~#
anager:~#
anager:~#
anager:~#
RBAC — Two Step Procell :-
① Create ClusterRole / Role
Namespace Specific
Cluster wide

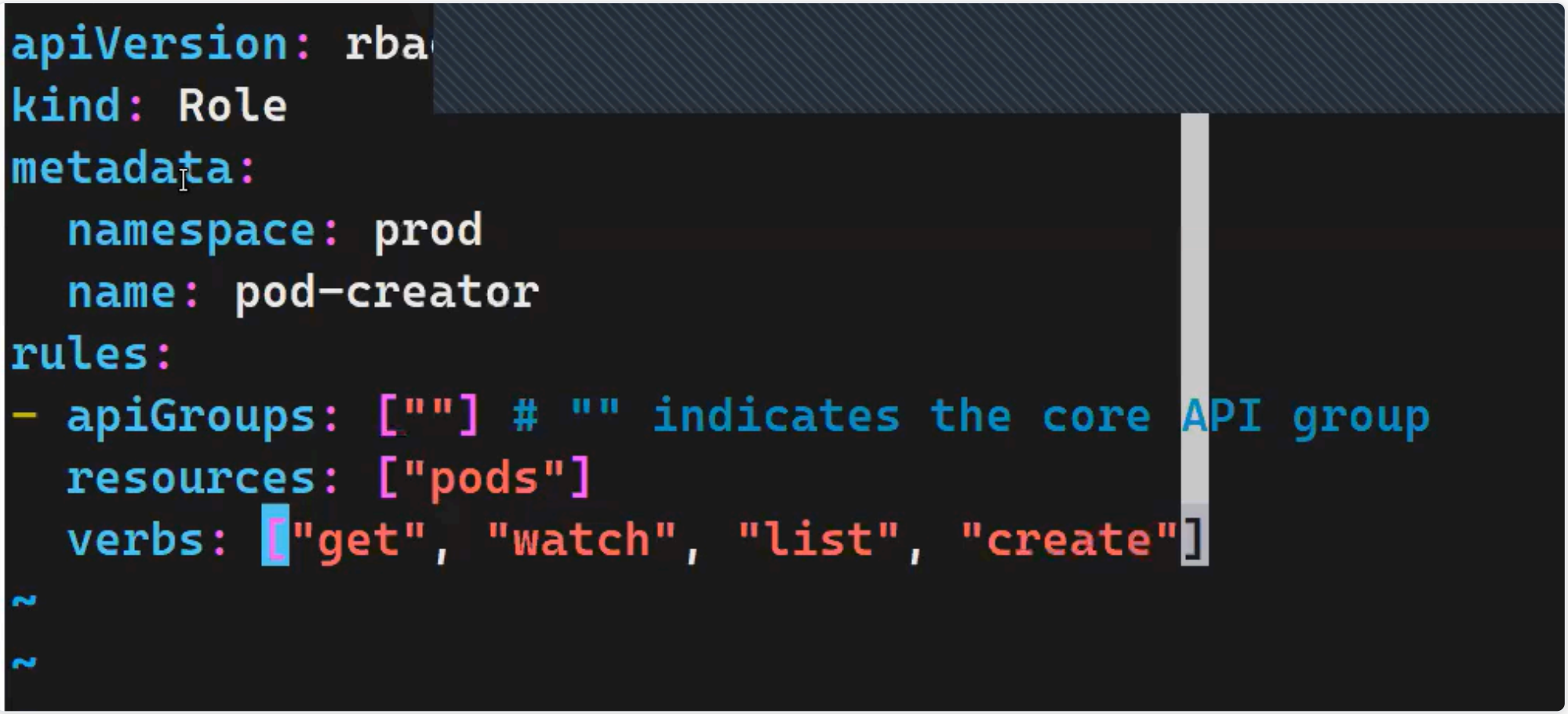

t@manager:~#
t@manager:~#
RBAC — Two Step Process :-
① Create ClusterRole / Role
Namespace Specific
Cluster wide
② Attach it to user.


alok@gitserver:~$
alok@gitserver:~$ openssl req -new -key ballu.key -out ballu.csr -subj "/CN=ballu"
alok@gitserver:~$
alok@gitserver:~$ ll ballu.*
-rw-rw-r-- 1 alok alok 875 Oct 7 12:09 ballu.csr
-rw------- 1 alok alok 1651 Oct 7 12:08 ballu.key
alok@gitserver:~$
alok@gitserver:~$ scp ballu.csr alok@10.0.0.100:
alok@10.0.0.100's password:
ballu.csr 100% 875 307.0KB/s 00:00
alok@gitserver:~$ scp ballu.key alok@10.0.0.100:
alok@10.0.0.100's password:
ballu.key 100% 1651 331.2KB/s 00:00
alok@gitserver:~$

```
root@manager:~# ll ballu.*
-rw-r--r-- 1 root root  981 Oct  7 12:11 ballu.crt
-rw-r--r-- 1 root root  875 Oct  7 12:10 ballu.csr
-rw------- 1 root root 1651 Oct  7 12:10 ballu.key
root@manager:~# kubectl get ns
NAME                 STATUS   AGE
default              Active   16d
kube-node-lease      Active   16d
kube-public          Active   16d
kube-system          Active   16d
monitoring           Active   2d23h
prod                 Active   120m
test                 Active   120m
```

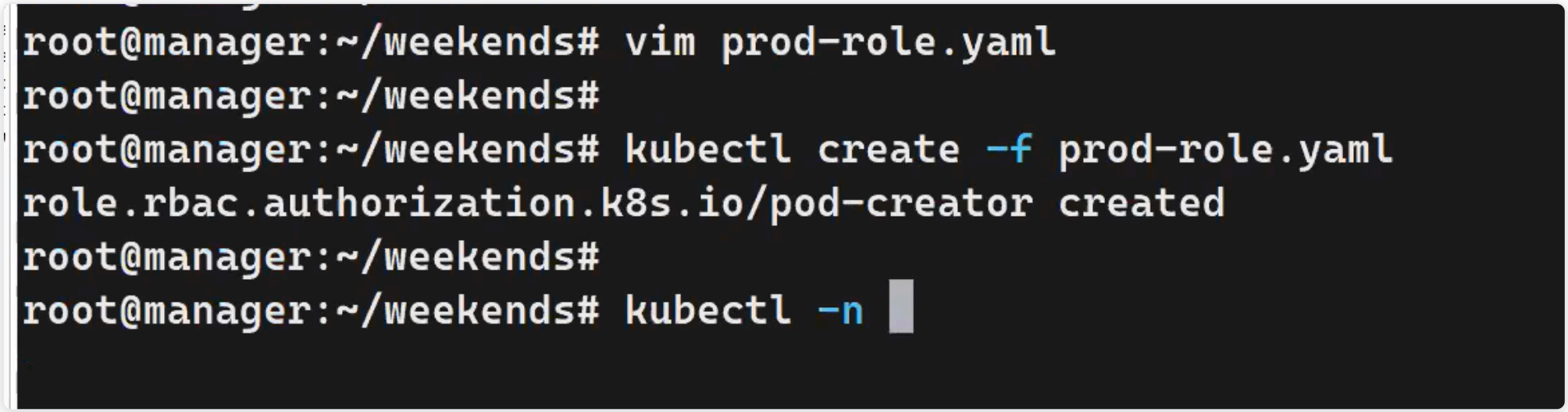
```
root@manager:~# kubectl config set-credentials ballu --client-certificate=ballu.crt --client
-key=ballu.key
User "ballu" set.
```

```
root@manager:~# kubectl config set-context ballu-context --cluster=kubernetes --namespace=pr
od --user=ballu
```

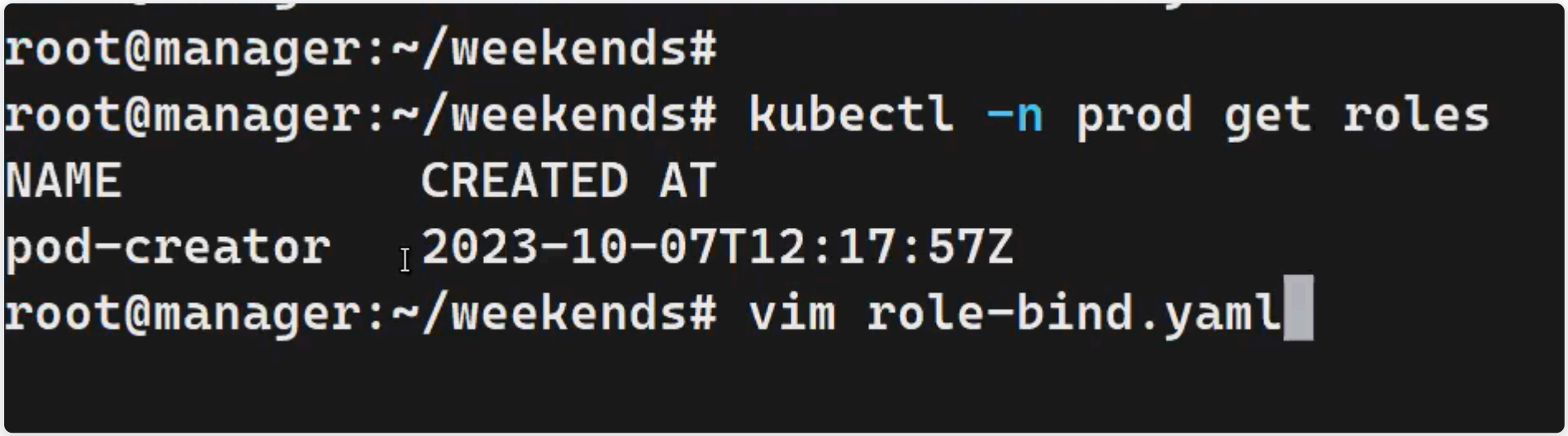

apiVersion: rba
kind: Role
metadata:
namespace: prod
name: pod-creator
rules:
- apiGroups: [""] # "" indicates the core API group
resources: ["pods"]
verbs: ["get", "watch", "list", "create"]
~
~

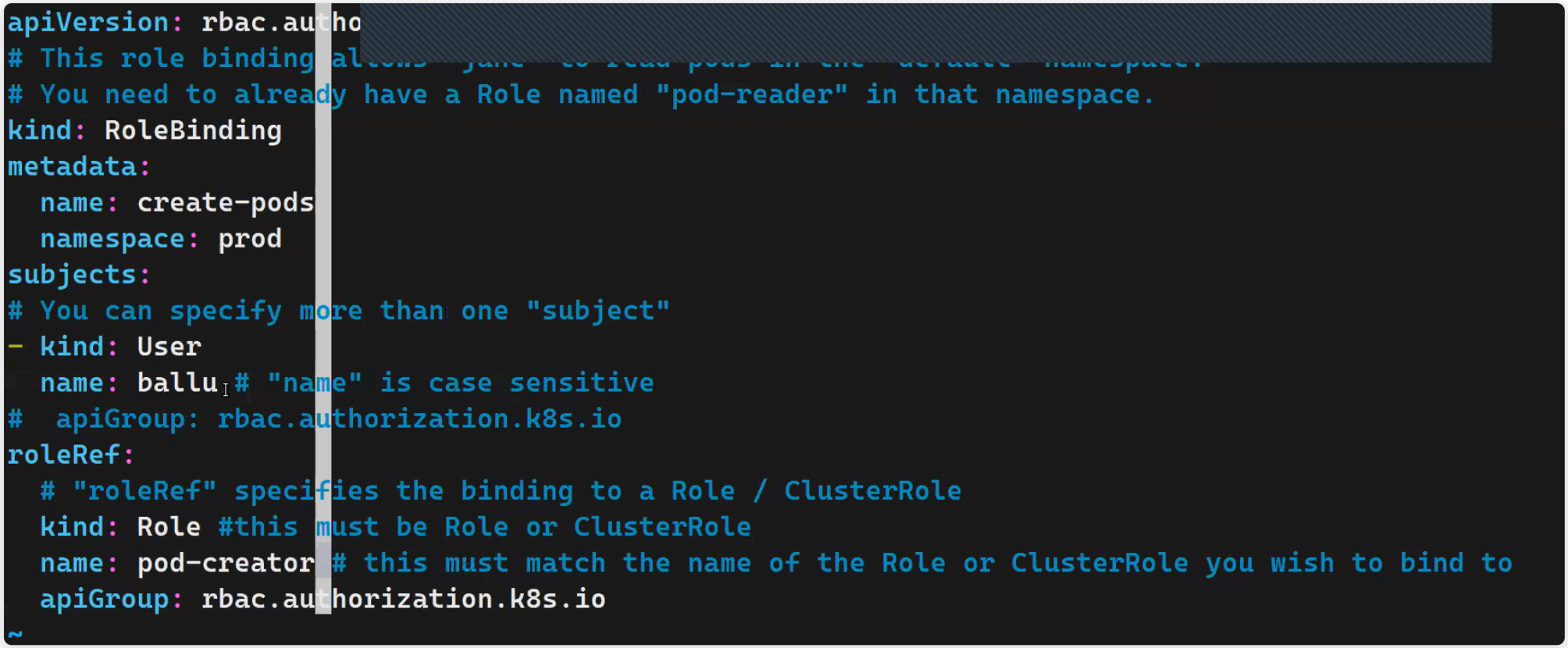

apiVersion: rbac.autho
kind: Role
metadata:
namespace: prod
name: pod-creator
rules:
- apiGroups: ["", "apps/v1"] # "" indicates the core API group
resources: ["pods", "deployments", "replicasets"]
verbs: ["get", "watch", "list", "create"]
~
~

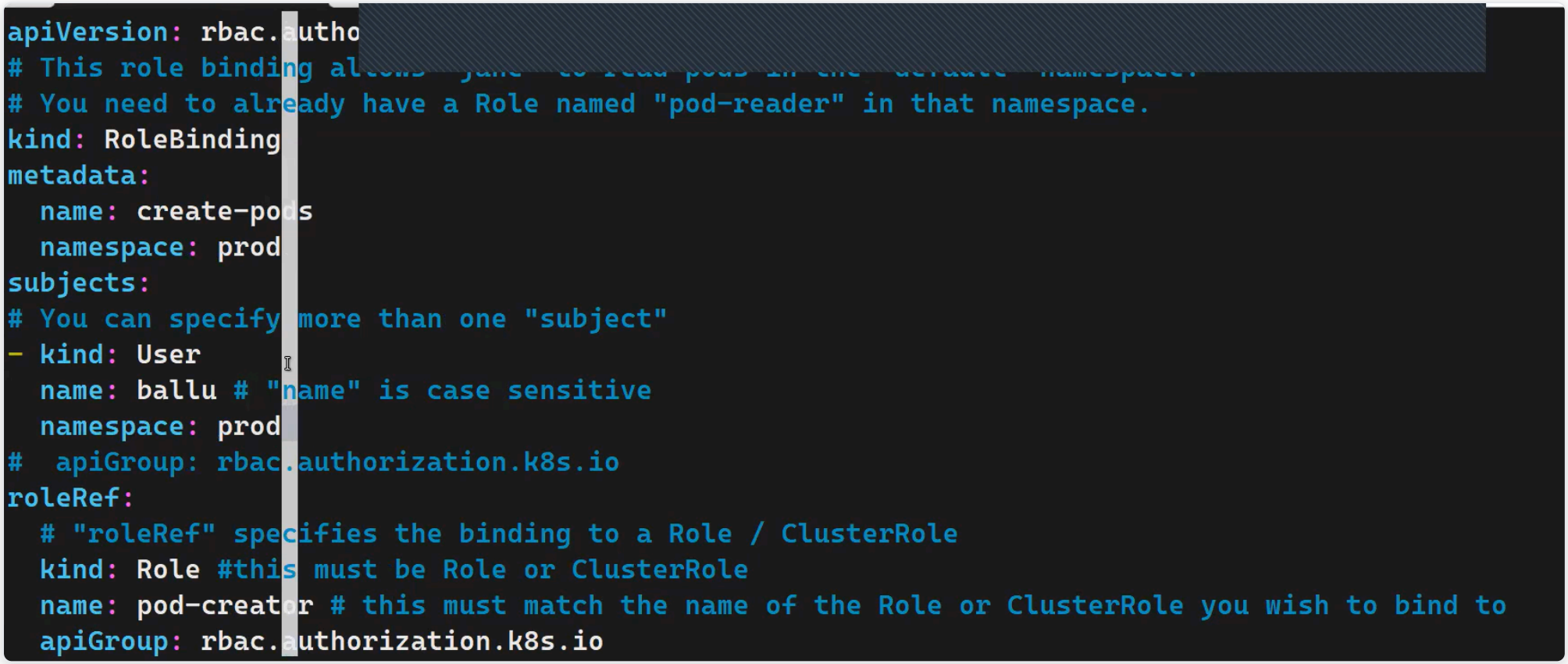
```
root@manager:~/weekends#
root@manager:~/weekends# vim prod-role.yaml
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f prod-role.yaml
```


root@manager:~/weekends# vim prod-role.yaml
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f prod-role.yaml
role.rbac.authorization.k8s.io/pod-creator created
root@manager:~/weekends#
root@manager:~/weekends# kubectl -n

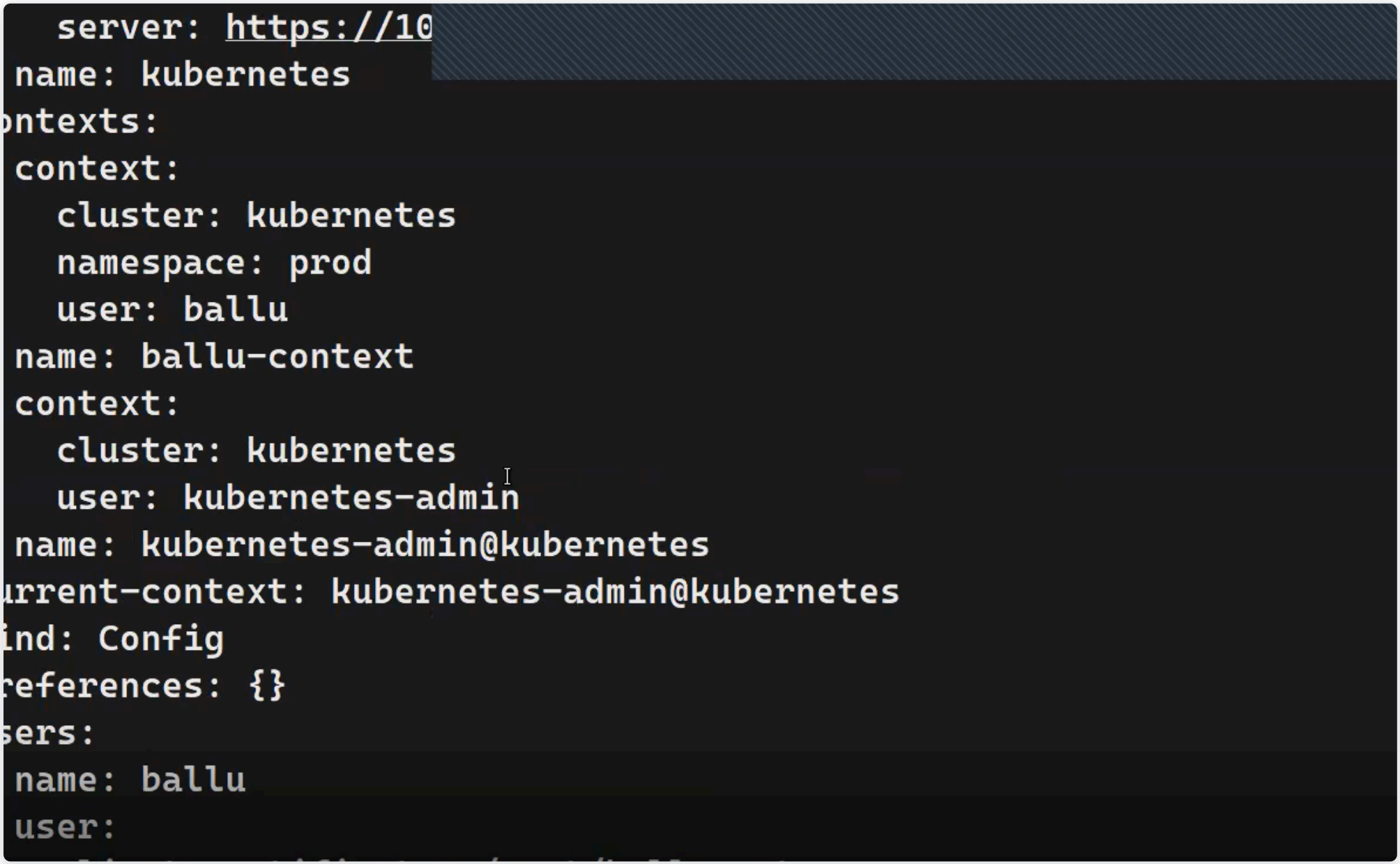
```
root@manager:~/weekends# vim prod-role.yaml
root@manager:~/weekends#
root@manager:~/weekends# kubectl create -f prod-role.yaml
role.rbac.authorization.k8s.io/pod-creator created
root@manager:~/weekends#
root@manager:~/weekends# kubectl -n prod describe role.rbac.authorization.k8s.io/pod-creato
Name:         pod-creator
Labels:       <none>
Annotations:  <none>
PolicyRule:
  Resources            Non-Resource URLs  Resource Names  Verbs
  ---------            -----------------  --------------  -----
  deployments          []                 []              [get watch list create]
  pods                 []                 []              [get watch list create]
  replicasets          []                 []              [get watch list create]
  deployments.apps/v1  []                 []              [get watch list create]
  pods.apps/v1         []                 []              [get watch list create]
  replicasets.apps/v1  []                 []              [get watch list create]
```

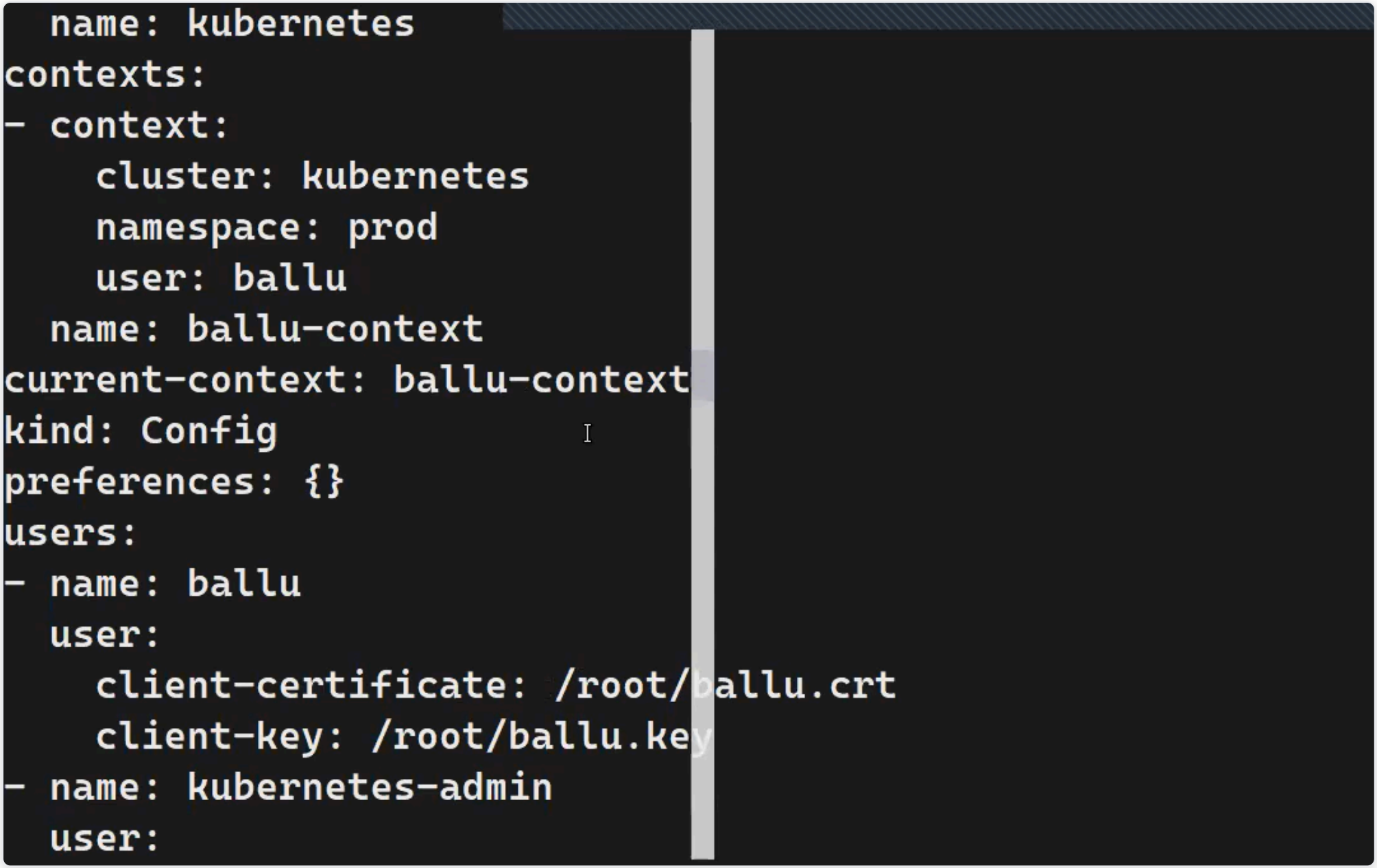

root@manager:~/weekends#
root@manager:~/weekends# kubectl -n prod get roles
NAME CREATED AT
pod-creator 2023-10-07T12:17:57Z
root@manager:~/weekends# vim role-bind.yaml

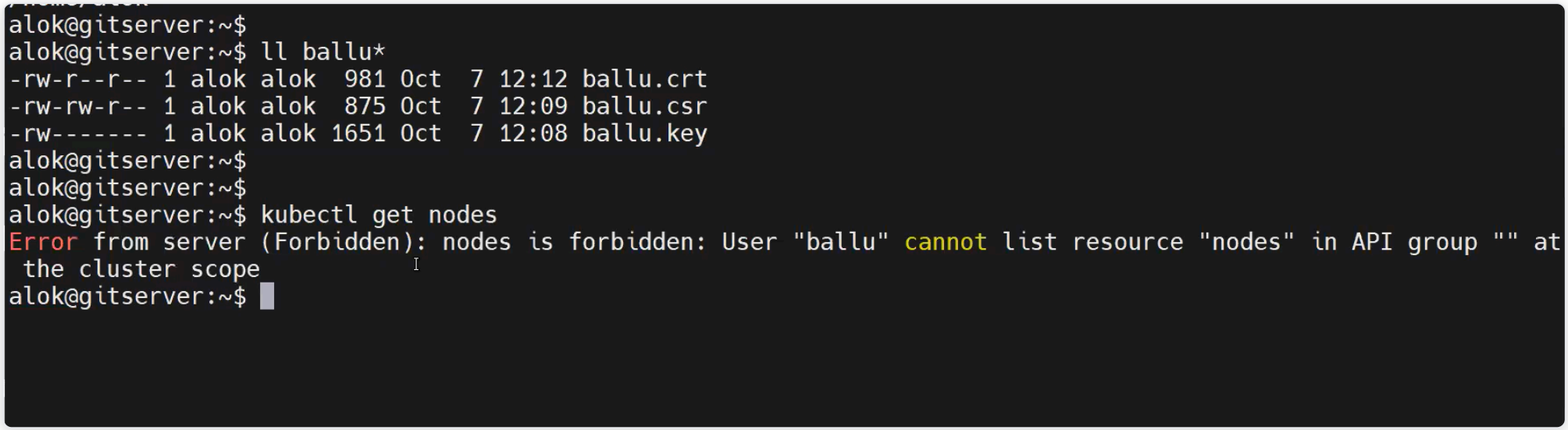

apiVersion: rbac.autho
This role binding allows "jane" to read pods in the "default" namespace.
You need to already have a Role named "pod-reader" in that namespace.
kind: RoleBinding
metadata:
name: create-pods
namespace: prod
subjects:
You can specify more than one "subject"
- kind: User
name: ballu # "name" is case sensitive
apiGroup: rbac.authorization.k8s.io
roleRef:
"roleRef" specifies the binding to a Role / ClusterRole
kind: Role #this must be Role or ClusterRole
name: pod-creator # this must match the name of the Role or ClusterRole you wish to bind to
apiGroup: rbac.authorization.k8s.io
~


apiVersion: rbac.autho
This role binding allows "jane" to read pods in the "default" namespace.
You need to already have a Role named "pod-reader" in that namespace.
kind: RoleBinding
metadata:
name: create-pods
namespace: prod
subjects:
You can specify more than one "subject"
- kind: User
name: ballu # "name" is case sensitive
namespace: prod
apiGroup: rbac.authorization.k8s.io
roleRef:
"roleRef" specifies the binding to a Role / ClusterRole
kind: Role #this must be Role or ClusterRole
name: pod-creator # this must match the name of the Role or ClusterRole you wish to bind to
apiGroup: rbac.authorization.k8s.io

```
root@manager:~/weekends# kubectl create -f role-bind.yaml
rolebinding.rbac.authorization.k8s.io/create-pods created
root@manager:~/weekends#
root@manager:~/weekends# kubectl -n prod describe rolebinding.rbac.authorization.k8s.io/create-pod
s
Name:         create-pods
Labels:       <none>
Annotations:  <none>
Role:
  Kind:  Role
  Name:  pod-creator
Subjects:
  Kind  Name   Namespace
  ----  ----   ---------
  User  ballu  prod
```


server: https://10
name: kubernetes
ontexts:
context:
cluster: kubernetes
namespace: prod
user: ballu
name: ballu-context
context:
cluster: kubernetes
user: kubernetes-admin
name: kubernetes-admin@kubernetes
urrent-context: kubernetes-admin@kubernetes
ind: Config
references: {}
sers:
name: ballu
user:


name: kubernetes
contexts:
- context:
cluster: kubernetes
namespace: prod
user: ballu
name: ballu-context
current-context: ballu-context
kind: Config
preferences: {}
users:
- name: ballu
user:
client-certificate: /root/ballu.crt
client-key: /root/ballu.key
- name: kubernetes-admin
user:


alok@gitserver:~$
alok@gitserver:~$ ll ballu*
-rw-r--r-- 1 alok alok 981 Oct 7 12:12 ballu.crt
-rw-rw-r-- 1 alok alok 875 Oct 7 12:09 ballu.csr
-rw------- 1 alok alok 1651 Oct 7 12:08 ballu.key
alok@gitserver:~$
alok@gitserver:~$
alok@gitserver:~$ kubectl get nodes
Error from server (Forbidden): nodes is forbidden: User "ballu" cannot list resource "nodes" in API group "" at
the cluster scope
alok@gitserver:~$

```
NAME       READY   STATUS              RESTARTS   AGE
ballupod   0/1     ContainerCreating   0          4s
burst      1/1     Terminating         0          129m
alok@gitserver:~$ kubectl get pods -n test
Error from server (Forbidden): pods is forbidden: User "ballu" cannot list resource "pods" in API group "" in t
he namespace "test"
alok@gitserver:~$
alok@gitserver:~$ kubectl get pods -n default
Error from server (Forbidden): pods is forbidden: User "ballu" cannot list resource "pods" in API group "" in t
he namespace "default"
alok@gitserver:~$
alok@gitserver:~$ kubectl get pods
NAME       READY   STATUS        RESTARTS   AGE
ballupod   1/1     Running       0          20s
burst      1/1     Terminating   0          129m
alok@gitserver:~$
alok@gitserver:~$ kubectl delete pod ballupod
Error from server (Forbidden): pods "ballupod" is forbidden: User "ballu" cannot delete resource "pods" in API
group "" in the namespace "prod"
alok@gitserver:~$
```

pods - CKA 5.00PM to 6.30PM (E) 2023-10-07 11-38-41 14th video 7th oct